Miter Bend Calculation Report

ASME B31.1-2022 | Pressure path and segment wall

DESIGN EQUATION

ASME B31.1-2022, 104.2.3(a)-(c)
Simplified paths: 10 psi or 100 psi + all applicable conditions
Otherwise: qualification under 104.7
(c)(3), close: ts = tm (2-r/R)/[2(1-r/R)]
(c)(3), wide: ts = tm [1 + 0.64 sqrt(r/ts) tan(theta)]
Segment thickness does not replace qualification.
Pressure equations: P and S in MPa; dimensions in mm; input bar x 0.1 = MPa.

Project	**QA miter**
Prepared by	**Device preview**
Report date	**2026-10-06 10:34**

01 INPUTS

Design gauge pressure	**8 bar**
Miter cut angle theta	**22.5 deg**
Nominal wall tn / crotch B	**6.02 / 40 mm**

02 DECLARED CONDITIONS AND PATH

(b)(1)-(2) θ ≤ 22.5° and every segment has B ≥ 6tn: Declared met

(a)(2) each segment thickness ≥ that required by para. 104.1: Declared met

(a)(3) fluid nonflammable, nontoxic and incompressible: Declared met

(a)(4) fewer than 7,000 full pressure cycles: Declared met

(a)(5) full penetration welds: Declared met

Selected path	**§104.2.3(c) — qualification per §104.7 required**
Pressure path limit	**Not a simplified path**

QUALIFICATION REQUIRED under 104.7. No approved pressure verdict is issued.

03 SEGMENT THICKNESS

Declared spacing	**CLOSE**
Straight-pipe minimum tm	**5 mm**
Pipe radius r	**50 mm**
Bend radius R	**150 mm**
Required segment wall ts	**6.25 mm**

104.2.3(c)(3) wall calculation does not replace qualification under 104.7. Geometry and spacing are designer declarations using ASME B31J.

04 REFERENCES AND LIMITS

Sources: B31.3-2018 304.2.3, supplied printed pp.22-23; B31.1-2022 104.2.3, supplied printed p.29. Inputs are user supplied. No external-pressure, fatigue, remaining-life or complete code qualification is performed.